

Uddipan Basu Bir

☎ +49 17667314674 | ✉ uddipanbb95@gmail.com | 🔗 LinkedIn | 🐙 GitHub | 📁 Portfolio | 📍 Erlangen, Germany

PROFESSIONAL SUMMARY

I am in the final phase of my M.Sc. in Data Science, specializing in AI and ML, with nearly five years of experience across Data Analytics, Data Engineering, and Applied AI. At Siemens, Schaeffler, and TCS, I have built end-to-end AI/ML solutions, production-ready RAG and NLP applications, ETL pipelines, dashboards, and automation workflows using Python, SQL, PySpark, Power BI, cloud platforms, and modern AI frameworks. I enjoy solving complex problems and building scalable solutions that run reliably in production.

TECHNICAL SKILLS

Machine Learning & AI: Deep Learning, Generative AI, RAG, Vision-Language Models, Multimodal Learning, Computer Vision, NLP, Time-Series Forecasting, Anomaly Detection, LoRA, QLoRA, PEFT

Programming & Frameworks: Python, SQL, PySpark, FastAPI, REST APIs, MCP, PyTorch, Scikit-learn, XGBoost, Hugging Face, LangChain, LangGraph, Google ADK

Data, Search & Knowledge Systems: Snowflake, dbt, pgvector, OpenSearch, Qdrant, FAISS, GraphDB, RDF, SPARQL, Cortex Analyst, Power BI

Cloud Platforms: AWS (Bedrock, Textract, ECS, ECR, Fargate), Azure (Azure ML, Azure OpenAI, Azure AI Search, App Service, Data Factory)

Data Engineering, MLOps & DevOps: Airflow, Kafka, Ab Initio, Autosys, Docker, Kubernetes, GitLab CI/CD, GitHub Actions, Jenkins, ArgoCD, MLflow, Langfuse, Prometheus, Grafana, TensorBoard

EXPERIENCE

Munich Music Labs, Technical University of Munich Germany

Munich, Germany

Research Team Member

Nov 2025 – Present

- Developing a multimodal audio-MIDI model for instrument assignment.

Siemens AG

Erlangen, Germany

Working Student – Applied AI Engineer

May 2025 – Present

- Built a FastAPI knowledge-graph dashboard with an LLM-powered natural-language-to-SPARQL engine using LangChain and GraphDB, deployed as containerized microservices on AWS Fargate.
- Built an end-to-end AI analytics platform on Siemens Data Cloud, using Cortex Analyst, Gemini/Google ADK, FastAPI, dbt, and Snowflake to generate governed SQL, executive reports, and chatbot-based insights.
- Developed and deployed Bidify, a RAG-based bid-analysis app, on AWS ECS/ECR with CI/CD, using Textract, Bedrock, and pgvector.
- Built four LlamaIndex tools for metadata-aware source and filter discovery in Siemens' production DQE RAG agent, integrating them with a GPT-4o ReAct agent and exposing them through an MCP server.

Tools & Technologies: FastAPI, LangChain, LlamaIndex, Google ADK, MCP, Snowflake, dbt, Cortex Analyst, GraphDB, OpenSearch, pgvector, AWS Bedrock, Textract, ECS/ECR, Fargate, Docker, GitLab CI/CD.

Schaeffler AG

Nürnberg, Germany

Working Student – Data Analytics and Machine Learning

Jun 2024 – Apr 2025

- Built an Azure ML pipeline using Isolation Forest and LightGBM to detect BOM price anomalies from SAP data and surface explainable alerts via APIs and Power BI.
- Developed an Azure ML capacity-forecasting solution using resource, project, and production data, deployed through batch and online endpoints for multi-level planning.

Tools & Technologies: Azure ML, Azure Data Factory, FastAPI, MLflow, Azure App Service, Power BI, SAP, Dataverse.

Tata Consultancy Services

Kolkata, India

Data Engineer / Data Analyst

Sep 2018 – Jun 2022

- Built and maintained 10+ production ETL workflows for BFSI data using Ab Initio, SQL, Unix, and Autosys, supporting reliable downstream analytics and regulatory reporting.
- Optimized large-scale data transformations by migrating processing logic from Ab Initio to PySpark, reducing runtime by 20%.
- Analyzed banking and financial datasets using SQL and Python, applying data cleaning, validation, aggregation, and trend analysis to support business reporting and decision-making.
- Developed Power BI dashboards and automated data-quality checks for financial KPIs, reducing manual effort by 20% and improving trust in reported data.

Tools & Technologies: Python, SQL, PySpark, Ab Initio, Autosys, Unix, Power BI, Kafka, Airflow, GitHub Enterprise, ServiceNow.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg

Erlangen, Germany

M.Sc. in Data Science, Major in Artificial Intelligence & Machine Learning

Oct 2022 – Present

- **Master Thesis:** Comparative analysis of vision-language models for OCR-based structured JSON extraction from document images.
- **Master Seminar Project (REHAU & FAU):** Forecasting and anomaly detection in time series data for polymer extrusion process. [GitHub](#)

Maulana Abul Kalam Azad University of Technology

Kolkata, India

Bachelor of Technology in Computer Science & Engineering

Jun 2014 – Jul 2018

- **Bachelor Thesis:** Emotion Detection Through Facial Expressions. [GitHub](#)

PROJECTS

Master Thesis: From Pixels to Structure | [GitHub](#)

- Fine-tuning and comparative analysis of multimodal and OCR-based vision-language models using PEFT methods such as LoRA, QLoRA, and Unsloth for structured JSON extraction from handwritten and printed document images.

Multi-modal, multi-view, multi-task Industrial Quality Inspection | [GitHub](#)

- Developed a multi-modal, multi-view, multi-task deep learning pipeline for classification and quality monitoring in industrial manufacturing.

Self-/Semi-Supervised Industrial Inspection Study | [GitHub](#)

- Built and compared self-supervised, semi-supervised, and combined deep learning approaches for image classification in industrial visual inspection of electrical motor manufacturing.

Local LLM-based RAG System for Personal Documents | [GitHub](#)

- Built a local, privacy-preserving RAG system for German bureaucracy letters using Streamlit, Ollama, OpenSearch hybrid retrieval, Redis caching, OCR-based ingestion, and Prometheus/Grafana observability, fully deployed via Docker.

Accident Severity Prediction from Sentiment Embeddings | [GitHub](#)

- Built a sentiment analysis pipeline on news reports to predict accident severity by comparing GloVe, Word2Vec, FastText, and Sentence Transformers embeddings with a Random Forest classifier.

HACKATHON

Agentic Fraud Detection Hackathon – [Reply](#) | [GitHub](#)

- Built a multimodal fraud-detection pipeline (transactions, SMS, email, audio) with rules-first triage and Langfuse-traced selective LLM review, cutting full-model processing to ~6.4% of cases.

LANGUAGES

English: Fluent **German:** Intermediate **Bengali:** Native **Hindi:** Fluent